Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Securing the Future with Quantum Computing and Cryptography

TGS Ref No: TGS-2024043419

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 12.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
10.0	1 February 2026	Previous release — 8-topic lesson plan with Topic 7 covering quantum paradoxes.	Dr Alfred Ang
11.0	11 August 2026	Realigned to the published TGS-2024043419 course outline: Topic 7 is now Quantum Computing Applications and Use Cases (finance, healthcare, science). Rebuilt the 4-day / 32-hour schedule around 18 hands-on activities mapped to the reusable Qiskit lab suite.	Dr Alfred Ang
12.0	11 August 2026	Major content expansion for the full 32 instructional hours: added a Prerequisites section (linear algebra, Dirac notation, unitary operators, density matrices and the five postulates of quantum mechanics), eleven full step-by-step mathematical derivations (phase kickback, Deutsch, Deutsch-Jozsa, teleportation, no-cloning, BB84 security, quantum error correction, Grover, Shor, CHSH and the QFT), Quantum Phase Estimation, and new sections on post-quantum hardening, quantum security and the convergence of quantum computing with AI. Integrated the QuantumLab Beginner ed-tool. Slide deck grows from 118 to 262 slides.	Dr Alfred Ang

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Course Information

Course Title	Securing the Future with Quantum Computing and Cryptography
WSQ Course Reference	TGS-2024043419
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	4 days · 8 training hours per day (32 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks counted within training time)
Mode	Instructor-led, hands-on quantum computing labs using Qiskit and the IBM Quantum Platform
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Design mathematical models required for quantum computing applications.
- LO2: Decide on appropriate physical methodologies specific to quantum computing problems.
- LO3: Direct two-level quantum models for quantum computing.
- LO4: Evaluate quantum computing applicability in the business context.
- LO5: Spearhead the application of quantum gates to quantum computing domains.
- LO6: Establish guidelines for selecting quantum computing algorithms.
- LO7: Synthesize quantum computing methodologies to address complex problems in quantum computing.

Target Audience and Prerequisites

- Target audience: quantum research scientists, quantum software developers, theoretical physicists, algorithm and cryptography researchers, computer science researchers, quantum hardware engineers, data scientists, postgraduate students in physics or computer science, and technology consultants.
- Knowledge and skills: computer operation proficiency, a minimum Polytechnic Diploma, and basic programming skill (preferably Python).
- Attitude: a positive learning disposition and enthusiastic engagement.
- Experience: a minimum of 1 year of working experience.
- Software and hardware: Python, Visual Studio Code or Anaconda, free Google Colab and IBM Quantum accounts, and a Windows or Mac laptop.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on quantum circuit and algorithm tasks, 1 hour, open book.

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- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted on Day 4 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- Learners assessed as Not Yet Competent may re-sit the assessment; an appeal process is available.

Training Delivery and Resources

- Delivery: instructor-led classroom or live virtual delivery, combining concept explanation, worked mathematical examples, live circuit demonstrations and hands-on activities.
- Every topic pairs concept teaching with at least one hands-on activity so learners apply each idea immediately.
- Learner resources: course slides, Learner Guide with full step-by-step activity instructions, and the lab source code (Python/Qiskit and OpenQASM) in the labs folder.
- Platform: Qiskit 2.x with the IBM Quantum Platform (free tier), IBM Quantum Composer, PennyLane and Google Colab.
- Trainer resources: master trainer slides, lesson plan, model answers and the marking guide.

Course Schedule

Each training day runs from 9:30 am to 6:30 pm and delivers exactly 8 instructional hours. The 1-hour lunch break is excluded from instructional time; tea breaks are counted within it.

Day 1 — Quantum Physics Fundamentals & Quantum Computing Systems

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, learning outcomes, ground rules and mandatory AM digital attendance (TRAQOM)
10:00–11:00	60 min	**PREREQUISITES — Mathematics of quantum mechanics: linear algebra and Dirac notation, inner/outer products, tensor products, eigenvalues. Unitary operators and the four properties (norm-preserving, inner-product preserving, reversible, unit-circle eigenvalues)**
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	**Topic 1 — The FIVE POSTULATES of quantum mechanics, one at a time with the mathematics of each (state space, unitary evolution, observables, Born rule, collapse). Density**

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		matrices and mixed states. Intro to the QuantumLab ed-tool
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Activity 1: Linear Algebra for Quantum Mechanics in Python
15:00–15:15	15 min	Tea break
15:15–16:15	60 min	**Topic 2 — Introduction to Quantum Computing Systems: qubits versus classical bits, the Bloch sphere, physical qubit technologies and commercial quantum platforms**
16:15–18:15	120 min	Topic 2 (continued) — Superposition and entanglement. Hands-on: Activity 2: Visualising a Qubit on the Bloch Sphere; Activity 3: Superposition and Entanglement on IBM Quantum
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total instructional time: 480 minutes (8 hours).

Day 2 — Quantum Gates, Circuits and Quantum Protocols

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory AM digital attendance (TRAQOM)
9:45–11:00	75 min	**Topic 3 — Quantum Gates and Circuits: Pauli gates, Hadamard, rotation and phase gates, and their key properties**
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	**Topic 3 (continued) — Multi-qubit gates (CNOT, CZ, controlled-phase, SWAP, Toffoli), Bell and GHZ states. FULL DERIVATION of PHASE KICKBACK, step by step, and the oracle identity U_f|x>|-> = (-1)^f(x)|x>|->**
12:30–13:30	60 min	Lunch break
13:30–15:15	105 min	Hands-on: Activity 4: Building Circuits in IBM Quantum Composer; Activity 5: Quantum Gates and Bell/GHZ States with PennyLane
15:15–15:30	15 min	Tea break
15:30–16:30	60 min	**Topic 4 — Quantum Protocols: FULL PROOF of the no-cloning theorem, and the complete seven-step**

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		algebraic DERIVATION of quantum teleportation. Superdense coding as its dual
16:30–18:15	105 min	Topic 4 (continued) — FULL PROOF of BB84 security: sifting, the intercept-resend attack, the 25% QBER and the detection probability. The CHSH inequality and its 2 vs 2√2 bound. Hands-on: Activity 6: Quantum Teleportation; Activity 7: Superdense Coding Protocol; Activity 8: BB84 Quantum Key Distribution with Eavesdropping Detection
18:15–18:30	15 min	Day 2 recap, Q&A and PM digital attendance

Total instructional time: 480 minutes (8 hours).

Day 3 — Quantum Algorithms and Quantum Cryptography

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory AM digital attendance (TRAQOM)
9:45–11:00	75 min	**Topic 5 — Quantum Algorithms: the query/oracle model. FULL DERIVATIONS of the Deutsch algorithm (6 steps) and Deutsch-Jozsa (5 steps) showing the exponential query separation**
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	Topic 5 (continued) — Bernstein-Vazirani. Hands-on: Activity 9: Deutsch and Deutsch-Jozsa Algorithms; Activity 10: Bernstein-Vazirani Algorithm
12:30–13:30	60 min	Lunch break
13:30–15:15	105 min	Topic 5 (continued) — FULL DERIVATION of Grover (two reflections = a rotation, the (π/4)√N bound and over-rotation). The QUANTUM FOURIER TRANSFORM and QUANTUM PHASE ESTIMATION. Hands-on: Activity 11: Grover's Search Algorithm; Activity 12: Quantum Fourier Transform
15:15–15:30	15 min	Tea break
15:30–16:45	75 min	**Topic 6 — Quantum Cryptography and Quantum Safety: classical cryptography, the quantum threat. FULL DERIVATION of Shor's algorithm (factoring**

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		reduced to period finding). POST-QUANTUM HARDENING: Mosca's inequality, cryptographic inventory (CBOM), NIST FIPS 203/204/205, hybrid key exchange and crypto-agility. QUANTUM SECURITY: threat model, QKD vs PQC, QRNG
16:45–18:15	90 min	Hands-on: Activity 13: Shor's Algorithm and the Quantum Threat to RSA; Activity 14: Post-Quantum Cryptography Migration Assessment
18:15–18:30	15 min	Day 3 recap, Q&A and PM digital attendance

Total instructional time: 480 minutes (8 hours).

Day 4 — Applications, Error Correction and Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory AM digital attendance (TRAQOM)
9:45–10:45	60 min	**Topic 7 — Applications and Use Cases: finance, healthcare and life sciences, science and materials, optimisation. QUANTUM COMPUTING AND AI: the four intersections, quantum machine learning honestly assessed, AI for quantum (calibration, ML error decoders), quantum-inspired classical algorithms and the realistic outlook**
10:45–11:00	15 min	Tea break
11:00–12:30	90 min	Hands-on: Activity 15: Molecular Simulation with VQE (Healthcare & Science); Activity 16: Portfolio Optimisation with QAOA (Finance)
12:30–13:30	60 min	Lunch break
13:30–14:30	60 min	**Topic 8 — Quantum Error Correction: FULL PROOF that QEC works — why naive copying is forbidden, encoding, syndrome operators and why measuring the syndrome does NOT collapse the logical state. Shor 9-qubit, Steane 7-qubit, surface codes and the threshold theorem**
14:30–15:45	75 min	Hands-on: Activity 17: Bit-Flip and Phase-Flip Error Correction Codes; Activity 18:

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		Shor 9-Qubit and Steane 7-Qubit Codes
15:45–16:00	15 min	Tea break
16:00–16:30	30 min	Course recap across all eight topics and Briefing for Assessment
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
17:30–18:30	60 min	**Practical Performance (PP) — hands-on quantum circuit and algorithm tasks, 1 hour, open book. TRAQOM survey and PM/Assessment digital attendance**

Total instructional time: 480 minutes (8 hours).

Topic and Activity Reference

Each topic maps to the learning outcomes shown and is taught with the hands-on activities listed. Detailed step-by-step instructions for every activity are in the Learner Guide.

Topic	Maps to	Hands-on activities	Slides
Topic 01: Quantum Physics Fundamentals	LO1	Activity 1	Slide 44
Topic 02: Introduction to Quantum Computing Systems	LO2, LO4	Activity 2, Activity 3	Slide 52
Topic 03: Quantum Gates and Circuits	LO3, LO5	Activity 4, Activity 5	Slide 63
Topic 04: Quantum Protocols	LO3, LO7	Activity 6, Activity 7, Activity 8	Slide 81
Topic 05: Quantum Algorithms	LO6, LO7	Activity 9, Activity 10, Activity 11, Activity 12	Slide 130
Topic 06: Quantum Cryptography and Quantum Safety	LO4, LO7	Activity 13, Activity 14	Slide 185
Topic 07: Quantum Computing Applications and Use Cases	LO4	Activity 15, Activity 16	Slide 216
Topic 08: Quantum Error Correction	LO7	Activity 17, Activity 18	Slide 234

Activity Reference

#	Activity	Learning outcome	Slides
1	Linear Algebra for Quantum Mechanics in Python	LO1	Slide 49
2	Visualising a Qubit on	LO2	Slide 58

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	the Bloch Sphere		
3	Superposition and Entanglement on IBM Quantum	LO2, LO4	Slide 60
4	Building Circuits in IBM Quantum Composer	LO5	Slide 76
5	Quantum Gates and Bell/GHZ States with PennyLane	LO3, LO5	Slide 78
6	Quantum Teleportation	LO3, LO7	Slide 123
7	Superdense Coding Protocol	LO3, LO7	Slide 125
8	BB84 Quantum Key Distribution with Eavesdropping Detection	LO7	Slide 127
9	Deutsch and Deutsch-Jozsa Algorithms	LO6	Slide 176
10	Bernstein-Vazirani Algorithm	LO6	Slide 178
11	Grover's Search Algorithm	LO6, LO7	Slide 180
12	Quantum Fourier Transform	LO6, LO7	Slide 182
13	Shor's Algorithm and the Quantum Threat to RSA	LO7	Slide 211
14	Post-Quantum Cryptography Migration Assessment	LO4, LO7	Slide 213
15	Molecular Simulation with VQE (Healthcare & Science)	LO4, LO7	Slide 229
16	Portfolio Optimisation with QAOA (Finance)	LO4, LO6	Slide 231
17	Bit-Flip and Phase-Flip Error Correction Codes	LO7	Slide 250
18	Shor 9-Qubit and Steane 7-Qubit Codes	LO7	Slide 252

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